Example 1.35

Solve the following system:

$x+2y+3z=0,\ 3x+4y+4z=0,\ 7x+10y+12z=0.$

Solution

Here the number of equations is equal to the number of unknowns.

Transforming into echelon form (Gaussian elimination method), the augmented matrix becomes

$$\left[\begin{array}{ccc|c}1&2&3&0\\3&4&4&0\\7&10&12&0\end{array}\right] \xrightarrow{\substack{R_2\to R_2-3R_1,\\ R_3\to R_3-7R_1}} \left[\begin{array}{ccc|c}1&2&3&0\\0&-2&-5&0\\0&-4&-9&0\end{array}\right] \xrightarrow{\substack{R_2\to R_2\div(-1),\\ R_3\to R_3\div(-1)}} \left[\begin{array}{ccc|c}1&2&3&0\\0&2&5&0\\0&4&9&0\end{array}\right]$$

$$\xrightarrow{R_3\to R_3-2R_2} \left[\begin{array}{ccc|c}1&2&3&0\\0&2&5&0\\0&0&-1&0\end{array}\right] \xrightarrow{R_3\to R_3\div(-1)} \left[\begin{array}{ccc|c}1&2&3&0\\0&2&5&0\\0&0&1&0\end{array}\right].$$

So, $\rho(A)=\rho([A\,|\,O])=3=$ Number of unknowns.

Hence, the system has a unique solution. Since $x=0,\ y=0,\ z=0$ is always a solution of the homogeneous system, the only solution is the trivial solution $x=0,\ y=0,\ z=0.$ ■

Note

In the above example, we find that

$$|A|=\begin{vmatrix}1&2&3\\3&4&4\\7&10&12\end{vmatrix}=1(48-40)-2(36-28)+3(30-28)=8-16+6=-2\neq 0.$$

Example 1.36

Solve the system: $x+3y-2z=0,\ 2x-y+4z=0,\ x-11y+14z=0.$

Solution

Here the number of unknowns is 3.

Transforming into echelon form (Gaussian elimination method), the augmented matrix becomes

$$\left[\begin{array}{ccc|c}1&3&-2&0\\2&-1&4&0\\1&-11&14&0\end{array}\right] \xrightarrow{\substack{R_2\to R_2-2R_1,\\ R_3\to R_3-R_1}} \left[\begin{array}{ccc|c}1&3&-2&0\\0&-7&8&0\\0&-14&16&0\end{array}\right] \xrightarrow{\substack{R_2\to R_2\div(-1),\\ R_3\to R_3\div(-2)}} \left[\begin{array}{ccc|c}1&3&-2&0\\0&7&-8&0\\0&7&-8&0\end{array}\right] \xrightarrow{R_3\to R_3-R_2} \left[\begin{array}{ccc|c}1&3&-2&0\\0&7&-8&0\\0&0&0&0\end{array}\right].$$

So, $\rho(A)=\rho([A\,|\,O])=2<3=$ Number of unknowns.

Hence, the system has a one parameter family of solutions.

Writing the equations using the echelon form, we get

$x+3y-2z=0, \qquad 7y-8z=0, \qquad 0=0.$

Taking $z=t,$ where t is an arbitrary real number, we get by back substitution,

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